

Box-Cox Lambda as a Redshift-Resolved Shape Statistic for Galaxy Colour Distributions: Advantages over Johnson SU Parameterisation

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This is a methods paper. It stands alone from the GAMA partial correlation paper but uses the same dataset as a demonstration case.

Abstract

We argue that the Box-Cox power parameter λ is a superior redshift-resolved shape statistic for galaxy colour distributions compared to the two shape parameters (a , b) of the Johnson SU family. The Johnson SU distribution is flexible and encompasses the Normal, log-Normal, and heavy-tailed distributions as limiting cases, but its two shape parameters are strongly correlated under maximum-likelihood optimisation, leading to numerical instability when fitting distributions that evolve smoothly across redshift bins. Box-Cox λ , by contrast, is a single interpretable parameter — the power that best normalises a positive distribution — with a direct physical meaning: $\lambda = 1$ implies additive (Normal) scatter, $\lambda \rightarrow 0$ implies multiplicative (log-Normal) scatter, and $\lambda < 0$ implies a distribution sharply bounded on one side, consistent with an absorbing state such as a consolidated red sequence. We demonstrate these properties using $u-r$ colour distributions for 165,488 galaxies from GAMA DR4 in seven redshift bins ($z = 0.002-0.35$). Box-Cox λ increases monotonically from -0.20 at $z = 0.03$ to $+2.70$ at $z = 0.33$ (Spearman $r = 1.000$, $p < 0.001$), while JSU parameters show large non-monotonic variation across the same bins despite the underlying distribution evolving smoothly. A bootstrap null test confirms the λ trend exceeds sampling noise at every bin. Python code implementing the full pipeline is provided.

1. Introduction

Parametric fitting of galaxy property distributions is a common tool for characterising populations and tracking evolution with redshift. For galaxy colours, the $u-r$ distribution is bimodal at low redshift, separating a red sequence from a blue cloud, but the degree of bimodality and the shape of each component evolve with cosmic time. Capturing this evolution requires a shape statistic that is (i) sensitive to the full distributional form, not just the mean; (ii) numerically stable across bins of varying size and shape; and (iii) physically interpretable.

The Johnson SU distribution (Johnson 1949) is a four-parameter family that transforms data to Normality via a hyperbolic sine function and is widely used in finance and engineering. Its two shape parameters a and b control skewness and kurtosis respectively. In principle it is well suited to galaxy colour distributions, which are unimodal-to-bimodal with varying tail

behaviour. In practice, we find that maximum-likelihood estimates of a and b are highly correlated — the likelihood surface is nearly flat along a ridge in (a, b) space — so small changes in the data can produce large jumps in the fitted parameters, even when the underlying distribution changes smoothly.

The Box-Cox power transform (Box & Cox 1964) is a single-parameter alternative. Given data $x > 0$, the transform $y = (x^\lambda - 1)/\lambda$ ($y = \log x$ when $\lambda = 0$) is chosen to make y as close to Normal as possible. The scalar λ is estimated by maximising the profile log-likelihood. Unlike the JSU optimisation, the Box-Cox likelihood surface is unimodal and well-behaved for the sample sizes typical of redshift surveys, and λ has a direct physical interpretation.

This paper makes the case for Box-Cox λ as a standard diagnostic in redshift-resolved distribution analysis, demonstrates the JSU instability on real data, and provides a bootstrap framework for assessing whether λ trends are physical or merely reflect sampling variation.

2. Statistical Background

2.1 Johnson SU Distribution

The Johnson SU family is defined by the transformation $z = a + b \sinh^{-1}((x - \xi)/\lambda)$, where $z \sim N(0,1)$. The four parameters are: location ξ , scale λ , and shape parameters a (skewness-related) and b (kurtosis-related, $b > 0$). Maximum-likelihood estimation of all four parameters simultaneously is sensitive to starting values and can converge to local optima when a and b are nearly collinear in the likelihood.

The key practical problem is parameter correlation. If the true distribution is close to symmetric, many (a, b) pairs fit equally well, and small perturbations in the data — including those introduced by moving from one redshift bin to the next — cause the optimiser to jump between equivalent solutions. This produces the artefact seen in Section 4: a changes sign and magnitude dramatically across adjacent bins while skewness (a direct moment-based measure) varies smoothly.

2.2 Box-Cox Power Transform

The Box-Cox transform is defined as:

$$y(\lambda) = (x^\lambda - 1) / \lambda \quad (\lambda \neq 0)$$

$$y(0) = \log(x) \quad (\lambda = 0, \text{ by continuity})$$

The parameter λ is estimated by maximising the profile log-likelihood of $y \sim N(\mu, \sigma^2)$ over λ . The likelihood surface for λ is smooth and unimodal for sample sizes $N \gg 100$.

Physical interpretation of λ for galaxy colour distributions:

- $\lambda = 1.0 \rightarrow$ approximately Normal; scatter is additive
- $\lambda \rightarrow 0 \rightarrow$ log-Normal; scatter is multiplicative
- $\lambda = -1 \rightarrow$ reciprocal transform best; distribution sharply bounded above
- $\lambda < 0 \rightarrow$ left tail suppressed; consistent with a one-way absorbing boundary

For $u-r$ colour, $\lambda < 0$ at low redshift corresponds to a consolidated red sequence acting as an absorbing state: galaxies quench and join the red sequence but do not reverse, producing a distribution sharply bounded on the red side with a long blue tail.

2.3 Bootstrap Null Test for Physical Evolution

To distinguish physical evolution from sampling noise, we construct a null distribution by bootstrapping: 500 subsamples of the same size as each redshift bin are drawn with replacement from the full galaxy pool (ignoring redshift), and λ is fitted to each subsample. The 2.5th and 97.5th percentiles form the 95% null envelope. If the observed λ lies outside this envelope, the variation is inconsistent with the null hypothesis that all bins are drawn from the same parent population. Monotonic trend is assessed with Spearman rank correlation and Kendall τ , both non-parametric.

3. Data

We use GAMA DR4 (Driver et al. 2022), specifically the StellarMassesLamdarv24 catalogue. After quality cuts ($0.5 < u-r < 4.0$, $\log M^* > 8.0$, $0.002 < z < 0.35$) the working sample contains 165,488 galaxies in seven equal-width redshift bins ($\Delta z = 0.05$; N per bin = 3,900–34,000). GAMA is well suited to this analysis: it is highly spectroscopically complete ($>98\%$ to $r < 19.8$), reducing photometric redshift scatter that would blur redshift bins.

[Cite GAMA DR4, stellar masses paper, spectroscopic completeness.]

4. Results

4.1 JSU Instability Demonstrated

Table 1 shows JSU parameters a and b alongside skewness and Box-Cox λ in each redshift bin. Skewness decreases smoothly from $+0.90$ to -0.82 — a coherent monotonic trend. Box-Cox λ increases smoothly from -0.20 to $+2.70$. JSU parameter a , however, jumps from -2.76 at $z = 0.026$ to -11.68 at $z = 0.075$, then to $+20.68$ at $z = 0.125$, before settling to values of order $3-10$ at higher redshift. This is not physical: the distributions at $z = 0.075$ and $z = 0.125$ differ only modestly in skewness (0.26 vs -0.15) and λ (0.43 vs 1.13). The instability arises because the JSU likelihood has a broad ridge in (a, b) space for near-symmetric distributions, causing the optimiser to find very different parameter pairs that are statistically equivalent but numerically distant.

[Insert Table 1: z -bin $\times$ (N , JSU a , JSU b , λ , skewness, kurtosis)]

[Insert Figure 1: side-by-side comparison of JSU a vs z (unstable) and λ vs z (smooth monotonic)]

4.2 Box-Cox λ Tracks Physical Evolution

Figure 2 shows Box-Cox λ and skewness against redshift with 95% bootstrap null envelopes. Both parameters lie outside the null envelope at every bin. Spearman $r = +1.000$ ($p = 4.0 \times 10^{-4}$) for λ and -1.000 ($p = 4.0 \times 10^{-4}$) for skewness, confirming monotonic physical evolution.

The λ trajectory is particularly informative. At $z = 0.325$, $\lambda = 2.70$: the distribution is broader than Normal toward the blue, reflecting a heterogeneous population of mostly star-forming galaxies with a growing red tail. By $z = 0.026$, $\lambda = -0.20$: the distribution is bounded sharply on the red side, the red sequence is consolidated, and the tail extends toward residual blue

galaxies still quenching. The crossing of $\lambda = 0$ (log-Normal regime) near $z \approx 0.08$ may correspond to the epoch at which the red sequence first becomes the dominant population — a testable prediction for deeper surveys.

[Insert Figure 2: λ and skewness vs z with bootstrap null envelopes]

5. Discussion

5.1 When to Use Box-Cox λ vs JSU

The JSU distribution remains useful when the goal is to generate synthetic realisations of a distribution or to compute tail probabilities analytically — tasks where a closed-form PDF is required. For tracking distributional evolution with a scalar statistic across redshift bins, Box-Cox λ is preferable: it is a single number, its likelihood surface is well-conditioned, and it has direct physical meaning. Skewness is an equally stable alternative but is sensitive to outliers and does not carry the same physical interpretation in terms of the power transform that normalises the data.

5.2 Generalisability

The argument is not specific to $u-r$ colour or to GAMA. Any redshift-binned analysis of a unimodal-to-mildly-bimodal distribution — stellar mass functions, star-formation rate distributions, metallicity distributions — could benefit from λ as a shape diagnostic. The bootstrap null test is entirely non-parametric and transfers directly.

One limitation: Box-Cox requires positive data. For distributions spanning zero (e.g., sSFR including quenched galaxies) a shift parameter is needed, introducing a choice that affects λ . In such cases, skewness remains the most assumption-free single-parameter shape statistic.

6. Conclusions

1. Johnson SU shape parameters a and b show large non-monotonic variation across redshift bins even when the underlying distribution evolves smoothly, due to correlation between parameters on a near-flat likelihood ridge.
2. Box-Cox λ is a single, well-conditioned, physically interpretable shape statistic that tracks the same evolution without numerical instability.
3. Applied to 165,488 GAMA galaxies in seven redshift bins, λ increases monotonically from -0.20 to $+2.70$ across $z = 0.03-0.33$ (Spearman $r = 1.000$, $p < 0.001$), consistent with progressive consolidation of the red sequence.
4. A bootstrap null test confirms the trend exceeds sampling noise at every redshift bin.
5. We recommend Box-Cox λ as the default shape statistic for redshift-resolved distribution analysis when a physically interpretable, numerically stable scalar is required.

Data and Code Availability

GAMA DR4 data: <https://www.gama-survey.org/dr4/>. Python analysis notebooks implementing the Box-Cox fitting and bootstrap null test are available at [GitHub URL to be

added]. Key functions require only `scipy.stats` (`boxcox`, `skew`, `kurtosis`) and `numpy`.